A LOCAL-FINITENESS OBSTRUCTION TO THE REGULARITY PREMISE IN FU'S THEOREM 2.8

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Abstract. In a 2026 manuscript investigating trans-exponential o-minimal expansions of the real ordered field, Theorem 2.8 assumes that for every allowed polynomial-exponential system P, the regular parameter locus $R \subset G \times [-1,1]^{2n+2}$ admits a triangulation by simplices $\{\Delta_\alpha\}$ that is locally finite in the full ambient parameter space $G \times [-1,1]^{2n+2}$. We prove that for the allowed system $n = 1,\ k = u = 0,\ P(x, l_1, l_2, \epsilon, \delta) = x^2$, the identity-coordinate slice of R in $[-1,1]$ is $[-1,1] \setminus \{0\}$, which is not closed. Because the union of any locally finite family of closed subsets in a topological space is closed, no locally finite triangulation of R by closed simplices exists in $G \times [-1,1]^{2n+2}$. If the simplices Δ_α are instead taken to be open simplex interiors, then the closedness step in the subsequent Claim 2.9 fails. The topological and logical core is verified in Lean 4.

1. Main Obstruction Theorem

Let $n = 1$ and $k = u = 0$, and consider the allowed polynomial system

$$P(x, l_1, l_2, \epsilon, \delta) \;=\; x^2.$$

For $(A, \eta, l, \epsilon, \delta) \in G \times [-1,1]^4$, the regular parameter set is defined by

$$R \;=\; \big\{(A, \eta, l, \epsilon, \delta) \in G \times [-1,1]^4 : P \circ A - \eta \text{ has nonsingular zeros only}\big\}.$$

Fixing the identity affine transformation $A = \mathrm{id}$ and auxiliary parameters $(l_1, l_2, \epsilon, \delta) = (0,0,0,0)$, the univariate polynomial $x^2 - \eta$ has a singular zero at $x = 0$ when $\eta = 0$ and only nonsingular real zeros (or no real zeros) when $\eta \in [-1,1] \setminus \{0\}$.

Theorem 1.1. *For $P(x, l_1, l_2, \epsilon, \delta) = x^2$, the slice of R along the identity coordinate $A = \mathrm{id}$ and $(l_1, l_2, \epsilon, \delta) = 0$ is*

$$R_{\mathrm{id}} \;=\; [-1,1] \setminus \{0\}.$$

Consequently:

(1) *R is not closed in $G \times [-1,1]^4$.*
(2) *There exists no family $\{\Delta_\alpha\}$ of closed simplices with $\bigcup_\alpha \Delta_\alpha = R$ that is locally finite at every point of $G \times [-1,1]^4$.*
(3) *If $\{\Delta_\alpha\}$ is instead a locally finite family of open simplices with $\bigcup_\alpha \Delta_\alpha = R$, then individual members Δ_α are not closed in $G \times [-1,1]^4$, obstructing the closed-membership inference in Claim 2.9.*

Proof. Since $0 \in \overline{[-1,1] \setminus \{0\}} \setminus R_{\mathrm{id}}$, R_{id} is not closed in $[-1,1]$, and by continuity of the slice inclusion $\eta \mapsto (\mathrm{id}, \eta, 0, 0, 0, 0)$, R is not closed in $G \times [-1,1]^4$. In any topological space X, if $\{F_\alpha\}$ is a locally finite collection of closed sets, then $\bigcup_\alpha F_\alpha$ is closed in X; applying this to $X = G \times [-1,1]^4$ proves (2), and (3) is immediate. □

Remark 1.2 (Scope Boundary)**.** Theorem 1.1 identifies a gap in the stated regularity premise of Theorem 2.8 and its application in Claim 2.9. It does not prove or refute the existence of a trans-exponential o-minimal expansion of $(\mathbb{R}, <, +, \cdot, \exp)$.

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